

Implementation of diagnostics services for primary health care in four Indian states



Ministry of Health & Family Welfare
Government of India



World Health
Organization
REGIONAL OFFICE FOR South-East Asia

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Contents

Foreword.....	v
Acknowledgements	vi
Abbreviations and acronyms	vii
Executive summary	ix
1. Introduction	1
Background	1
Methodology.....	2
2. Select IVD models.....	5
Overview	5
Telangana	8
Tamil Nadu	14
Chhattisgarh.....	19
Odisha.....	24
Salient features of the IVD models.....	27
Pros and cons of different IVD models	29
Conclusion – IVD models	30
3. Teleradiology models.....	31
Teleradiology	31
Case study: Telangana	32
4. Conclusion.....	36

Foreword

Limited access to clinical diagnostics, particularly at the primary care level, impedes the pursuit of universal health coverage. Although diagnostic information is required for high-quality provision of therapeutic care, diagnostics are chronically under-prioritized. The WHO Model List of Essential Diagnostics was created to easily raise prioritization of diagnostics. Some Member States have also launched the creation of their own national essential diagnostics lists. The first country to do this was India, publishing their list in 2019.

As a fairly new approach to improving access to diagnostics, the experiences in implementing a national essential diagnostics list are few. This technical report provides a snapshot of such implementation, examining how four states in India improved access to in vitro diagnostics at the primary health care level, via sample transport systems. It also examines how teleradiology has been implemented to extend access to various modes of diagnostic imaging down to primary health care facilities.

As detailed in the following pages, implementation of a national essential diagnostics list has resulted in significant improvement in access to diagnostics, including both in vitro diagnostics and other tests such as radiological and ultrasound imaging. The methods used to establish these services across different states in India have both common features as well as unique aspects. Other WHO Member States can learn from the service models described herein to improve access to diagnostics services for their own populations.



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Abbreviations and acronyms

AIIMS	All India Institute of Medical Science
CGMSC	Chhattisgarh Medical Services Corporation
CHC	community health centre
CR	computed radiography
CT	computed tomography
DH	district hospital
DPHL	District Public Health Laboratory
ELCOT	Electronics Corporation of Tamil Nadu
EQAS	External Quality Assurance Services
FDSI	Free Diagnostics Service Initiative
HL7	Health Level Seven
HMIS	Health Management Information System
HR	human resources
IELS	Integrated Essential Laboratory Services
IQC	internal quality control
IT	information technology
IVD	in-vitro diagnostics
ISO	International Organization for Standardization
LMIS	Laboratory Management Information System
MCH	Medical College Hospital
MoHFW	Ministry of Health & Family Welfare
MRI	magnetic resonance imaging
NEDL	National Essential Diagnostics List
NHSRC	National Health Systems Resource Centre
OSMC	Odisha State Medical Corporation
PACS	Picture Archiving and Communication System
PHC	primary health centre
POC	point-of-care
PPP	public–private partnership
SDH	sub-district hospital
SEARO	WHO Regional Office for South-East Asia
SHC	Sub-health centre
SOP	standard operating procedure

TB	tuberculosis
TNMSC	Tamil Nadu Medical Services Corporation
TSMSIDC	Telangana State Medical Services and Infrastructure Development Corporation
UHC	universal health coverage
USG	ultrasonography
WHO	World Health Organization

Executive summary

Only a few countries in the world have completed the development of a National Essential Diagnostics List (NEDL), based upon the WHO Model List of Essential Diagnostics (first version published in 2018). India completed and published its NEDL in 2019. As a relatively new concept, questions have arisen. “Can implementation of diagnostics services based on the NEDL make a significant impact in low-resource settings? What would implementation look like?”

With responsibility for health-care services residing at the state level in India, diagnostics services implementation has taken different trajectories, with either partial or full compliance with the India NEDL in various states. The project described herein examined the NEDL implementation experience among different states in India, in an attempt to describe the main elements of implementation and identify factors for success. This report describes the in vitro diagnostics (IVD) models in four Indian states: Telangana, Tamil Nadu, Chhattisgarh and Odisha. Their setups and salient features are described, with pros and cons observed among the different ways of operating. Enablers for success are also described. The Government of India published a guidance document for free laboratory services that became the operational guidelines for implementing diagnostic services in a hub and spoke model. These guidelines covering technical details and case studies from state governments are an important national-level enabler.

Results showed that NEDL implementation significantly improves access to diagnostics. The state of Telangana saw a large increase in IVD test utilization over four years, from 3.6 million tests at one hub laboratory in 2018 prior to NEDL implementation, to 52 million tests in 31 hub laboratories in 2022. In the state of Chhattisgarh, where implementation started only in 2020, a large district hospital testing hub saw a threefold increase in IVD test utilization from 33 000 tests per month in 2020 to 100 000 tests per month in 2023. All four states established access to IVD tests at facilities without laboratories. IVD testing went from zero to between five and 20 IVD tests per day per primary health-care facility, enabled via sample transport to hub testing facilities. Across the four states examined, the range of IVD tests available is 134–146 tests. Results are available within 12–24 hours for the majority of tests, with results for some high-end tests taking one to eight days, as per the turnaround time defined by the state.

Teleradiology services have been established in 11 states in India, providing access to imaging services at lower-level facilities that lack a specialist such as a radiologist for image analysis. Teleradiology services for X-rays includes digitization, transmission and reporting by a radiologist. The services also include computed tomography (CT) scan and magnetic resonance imaging (MRI) at higher health facilities.

Although beyond the scope of this study, the dramatically improved access to and utilization of IVD and imaging tests has certainly improved health outcomes and reduced out-of-pocket expenditure. Other positive effects can be assumed, such as reduced antimicrobial resistance and gains in efficiencies by pushing health-care service utilization down to lower-level facilities.

1. Introduction

Background

Universal health coverage (UHC) can be achieved only with an effective primary health care system that provides accessible, affordable and equitable health-care services to all. Diagnostics play a vital role in establishing high-quality primary health care services. About half of the global population has limited access to diagnostics services, with about four in five people in low-income and lower-middle-income countries lacking access to the simplest of diagnostic tests (1). We cannot achieve UHC, disease surveillance and control, and antimicrobial resistance mitigation without strengthening diagnostic services. Populations require much better access to in-vitro diagnostic (IVD) tests as well as other diagnostic services, such as imaging.

Providing diagnostic services for population-wide access requires strengthening or establishing new system components, such as specimen transportation services for IVDs, streamlined procurement and supply chain mechanisms, new digital tools, and extending operational functions and resources to reach lower facility levels, such as information technology (IT) systems, human resources (HR). Due to human and financial resource limitations, existing gaps in IVD testing access can be addressed with effective sample transportation from remotely located areas and utilization of IT systems for reporting and for supporting quality assurance.

Teleradiology enables people to receive imaging services in locations that lack a specialist for image interpretation and analysis, such as a radiologist. Via transmission of radiographic images and reporting through IT systems that include picture archiving and communication system (PACS) software, services become accessible in lower-level health-care facilities.

To address high out-of-pocket expenditure arising from poor access to diagnostics, the Ministry of Health & Family Welfare (MoHFW), Government of India under the banner of the National Health Mission launched the Free Diagnostics Service Initiative (FDSI) in July 2015. The Initiative was envisioned to provide a set of essential diagnostics at different levels of health-care service delivery to enable caregivers to make proper decisions regarding the diagnosis and treatment of patients. In the FDSI, the Central Government supports the states to provide a set of essential diagnostics, including both laboratory and radiology tests at their health-care facilities free of charge to beneficiaries.

In 2018, WHO published its first Model List of Essential Diagnostics, which countries can use to develop their own national lists. The fourth version (2) was published in October 2023. India developed and published the world's first National Essential Diagnostics List (NEDL) in 2019, which built upon the initial list created with the FDSI guidelines. Two other countries in the WHO South-East Asia Region have launched development of their NEDLs, Nepal and Timor-Leste. Nepal endorsed and published its list in early 2024. Several more countries globally are planning to develop or have already started developing their NEDLs. Because India has several years' experience in implementing

1 The Lancet Commission on diagnostics: transforming access to diagnostics. Lancet. 2021;398(10315):1997–2050. [https://doi.org/10.1016/S0140-6736\(21\)00673-5](https://doi.org/10.1016/S0140-6736(21)00673-5).

2 The selection and use of essential in vitro diagnostics: report of the fourth meeting of the WHO Strategic Advisory Group of Experts on In Vitro Diagnostics, 2022 (including the fourth WHO model list of essential in vitro diagnostics). Geneva: World Health Organization; 2023 (WHO Technical Report Series, No. 1053). Licence: CC BY-NC-SA 3.0 IGO.

its NEDL, this project was conducted to capture key features and identify factors for success, from IVD and teleradiology models functioning at the primary health care level across different states in India.

Primary health care facilities have the greatest limitations in availability of specialized HR, diagnostic equipment and other resources, yet they are the most accessible to the majority of the population. Although these facilities either lack laboratories or have limited testing capabilities, access to a large range of IVD services can be established through specimen transport systems. Thus, the study's scope for examination of IVD services in India focused on select states that have established access at primary health care facilities via sample transport.

Health-care facilities where teleradiology can be accessed requires imaging equipment such as X-ray and computed radiography (CR) systems and skilled technicians (radiographers) for their operation. This is typically at levels higher than primary care clinics. The scope of examination of teleradiology services focused on the lowest level where imaging occurs, with image analysis conducted remotely at another facility.

Methodology

A Technical Advisory Group was assembled to oversee the study. This included members from the WHO Regional Office for South-East (SE) Asia, WHO India Country Office and the National Health Systems Resource Centre (NHSRC), Division of Healthcare Technology and Innovations of India, MoHFW. NHSRC is a WHO Collaborating Centre for priority medical devices and health technology policy. Their approved workplan includes provision of technical inputs that may inform WHO's work on assessments of the use of IVDs and medical imaging. NHSRC is a premier think-tank for MoHFW, providing technical assistance in policy-making and its implementation and uptake in the states and union territories of India. NHSRC was instrumental in providing guidance and facilitating investigations of the service models in the target states.

Fig. 1. Methodology

1 Planning	2 Field visits and assessment	3 Analysis and reporting
Key activities		
• Inception meeting • Desk review • Identification of IVD and teleradiology models (long list) • Shortlisting of IVD and teleradiology models for primary assessment • Primary assessment tools and finalization of assessment plan • Formation of Technical Advisory Committee	• Conducting field visits • Interviews with stakeholders • Collating responses and their analysis • Presentation on findings from the field	• Analysis of each implementation • Identification of best practices and success factors • Drafting report • Presentation of draft findings • Finalizing report
Outputs		
• Identification of models of IVD and teleradiology services for primary assessment • Final assessment plan	• Data collection and compilation • Brief report – IVD and teleradiology	• Interim report • Final report

A phased approach was adopted for the study, with three phases, represented in Fig. 1. During the planning phase, a desk review was done to arrive at the IVD and teleradiology models present at the primary health care level in India. The models were longlisted based on multiple parameters. The four states of Telangana, Tamil Nadu, Chhattisgarh and Odisha were selected for further study with field visits and assessments, based on discussions with the Technical Advisory Group.

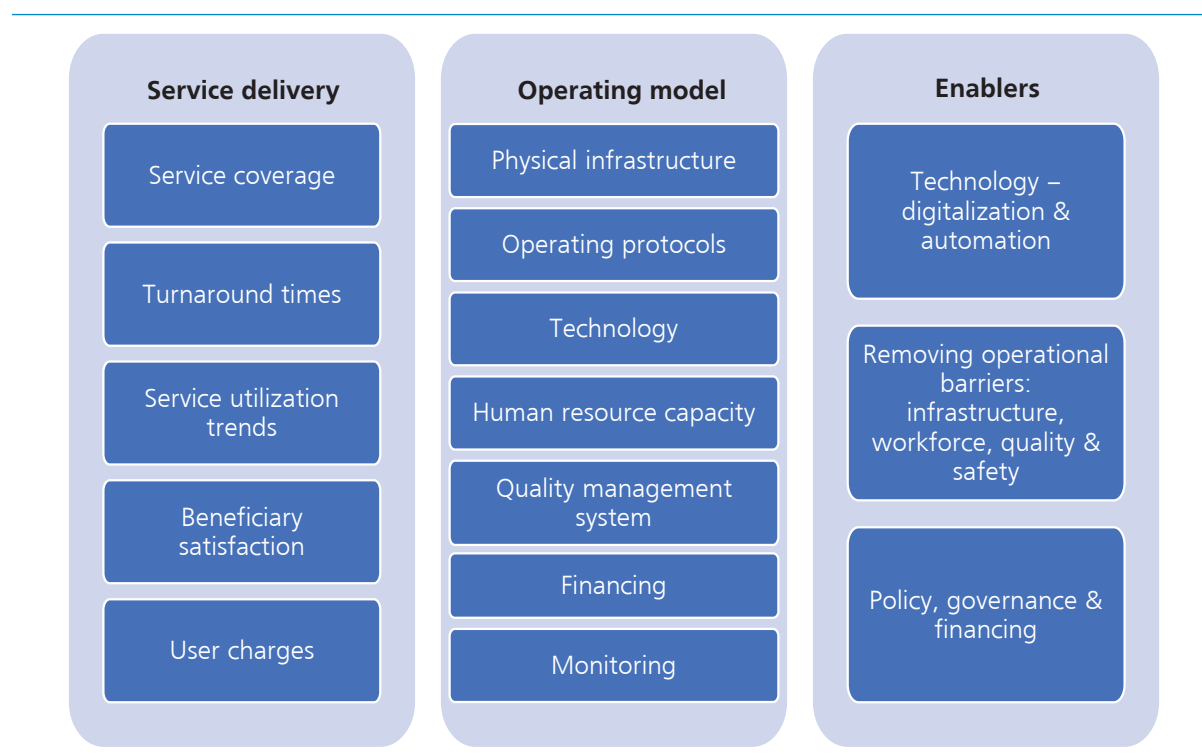
Field visits were conducted per the plan below.

Table 1. Field visit plan

#	State	Field visit date
1	Telangana	1/11/23–3/11/23
2	Tamil Nadu	7/11/23–10/11/23
3	Chhattisgarh	21/11/23–24/11/23
4	Odisha	21/11/23–23/11/23

The methodology used during the field visits were observations, in-depth interviews with stakeholders, discussions with facility staff and discussion with beneficiaries. Assessments for the facilities visited were carried out based on NHSRC's Checklist–Self Assessment of Free Diagnostic Services Initiative Tool. The observations and inputs received during the field visits on the IVD and teleradiology models were analysed based on the three major aspects of service delivery, operating model and enablers. The components under each major aspect are represented in Fig. 2.

Fig. 2. Aspects of analysis



For the service delivery component, analysis was done of service coverage of the model, turnaround time, utilization trends of the services, satisfaction of beneficiaries and user charges for delivering the services.

In the operating model component, the physical infrastructure, including technology present at different levels for delivering services, was analysed along with HR capacity. Operating protocols, quality and safety parameters were also analysed. Financing and monitoring of services at various levels were analysed. Information was obtained on financing and utilization based on discussions with stakeholders from the respective states.

For each state model, based on the initial analysis of service delivery and operating model, apparent enablers for success were identified, Enablers that could push models to perform well were further identified from factors such as technology, infrastructure, workforce and quality. The policy and governance mechanisms that enable the model to perform were also examined.

Terminology for facility types in India is shown below, as described in the India NEDL (3) in order from the highest level of care to the lowest.

DH = district hospital

Above the DH level are specialist hospitals, such as apex medical college hospital (MCH) or ultra-apex MCH.

SDH = subdistrict hospital

CHC = community health centre, basic hospital

PHC = primary health centre

SHC = subhealth centre

Note that this method of interviewing various stakeholders has provided estimated figures for the number of types of tests offered at each facility level and total number of samples tested in each state, at the time of the interviews. Because implementation in each state is an ongoing and dynamic situation, these figures should be taken as estimates and not official government data. However, the estimated figures are useful for informing readers in the SE Asia Region and beyond, that implementation of a country's NEDL is achievable and significantly increases utilization of diagnostics services.

3 India National Essential Diagnostics List. New Delhi: Indian Council of Medical Research 2019 (https://main.icmr.nic.in/sites/default/files/guidelines/NEDL_2019.pdf, accessed 12 July 2024)

2. Select IVD models

IVD models were studied in four states, Telangana, Tamil Nadu, Chhattisgarh and Odisha. These states have used either an in-house model or a hybrid model that includes public–private partnership (PPP). An overview of the characteristics is given in the three tables below for the major aspects: service delivery, operating model and enablers. A more detailed review of the models in each of the states is provided after the overview.

Overview

A brief snapshot of service delivery characteristics across the states is shown in Table 2.

Table 2. Analysis of IVD models through the service delivery component

Service delivery					
#	Parameter	Telangana	Tamil Nadu	Chhattisgarh	Odisha
1	Service coverage	Total 134 tests	DH level: 134 tests Apex MCH: 182 tests Ultra-apex MCH: 255 tests	District-level hub: 114 tests CHC-level hub: 54 tests	Total 146 tests Out of 146 tests, 52 high-end tests were provided through PPP mode
		Operational with 31 DH hubs	Piloted hubs in one district	Operational with hubs set up in one MCH, 13 DHs, seven CHCs	Operational with hubs in DH/SDH and CHCs, connected to six PPP regional hubs
2	Sample transport	Outsourced, via dedicated vehicles from a vendor	In-house, via tuberculosis (TB) programme senior treatment supervisor, using own transportation	In-house, via TB runners	In-house from PHCs to CHCs At higher levels, via PPP service provider
3	Turnaround time*	95% of test results available within 12–24 h	Approximately 90% of test results available within 12–24 h	Approximately 90% of test results available within 24 h	In-house: most within 24–48 h; Outsourced high-end tests: from 1 to 8 days
4	Service utilization trends	52 million tests for all hubs across the state (2022), from 3.6 million tests (2018) for one pilot hub About 10–15 samples/day/spoke**	About 10–20 samples/day/spoke** during pilot phase	Over 70 000 tests per month (early-to mid-2023), from 20 800 tests per month (2020) at one DH hub About 5–10 samples/day/spoke**	About 9000 tests per day at one of the regional labs About 10–15 samples/day/spoke**

Service delivery					
#	Parameter	Telangana	Tamil Nadu	Chhattisgarh	Odisha
5	Beneficiary satisfaction	High			
6	User charges	Free of charge to all beneficiaries			

* FDSI guidelines provide the recommended total (including reporting) turnaround times for each test, which varies from less than an hour (e.g. rapid tests) to eight days (e.g. antinuclear antibody).

** In the table above, the term “spoke” is the lowest-level facility that transports specimens to a hub diagnostics laboratory, described under each state in the following sections.

The operating models had similarities as well as distinct variations. Odisha was the only state that had a hybrid model, where a certain subset of tests was being provided through a PPP model. All the models have been strengthening in-house capacities to offer an enhanced range of services, establishing standard operating procedures (SOPs) and utilizing technology to varying degrees (Table 3).

Table 3. Analysis of IVD models through the operating model component

Operating model					
#	Parameter	Telangana	Tamil Nadu	Chhattisgarh	Odisha
1	Physical infrastructure	District hub labs (new and separate within DH complex) in each district Cold boxes at spokes for sample transportation	Central lab at MCH acts as apex lab of district CHC, SDH, DH act as hub labs and spokes Cold boxes at spokes for sample transportation	District and CHC hub labs established within existing health facilities building Cold boxes at spokes for sample transportation	In-house labs strengthened in PHC, CHC and DH Six regional labs set up by PPP service provider for outsourced tests Cold boxes at spokes for sample transportation provided by the PPP service provider
2	Operating protocols	SOPs are defined at each level			
3	Technology Automated diagnostic equipment	Laboratory Management Information System (LMIS) to capture entire process, sample barcoding at spokes Analysers, microscope, rapid diagnostic tests, etc.	LMIS software, integrated essential laboratory services web and mobile applications for sample transport, with barcoding of samples integrated with LMIS Analysers, microscope, rapid diagnostic tests, etc.	LMIS and barcoding of samples at district hubs Analysers, microscope, rapid diagnostic tests, etc.	LMIS including barcoding of samples for outsourced tests Analysers, microscope, rapid diagnostic tests, etc.

Operating model					
#	Parameter	Telangana	Tamil Nadu	Chhattisgarh	Odisha
4	HR capacity	Adequate staffing ensured at hub labs and spokes Resources also pooled from other programmes			
5	Quality management system	Internal quality checks run daily on the equipment Established mechanism for external quality assessments			
6	Financing	Fully funded by the government			
7	Monitoring	Defined system of monitoring and reviews at district and state levels			

Service delivery and implementation of the operating model were aided by enablers, in which technology has played a vital role, the government has worked towards removing operational barriers, and there has been strong political and administrative will to deliver high-quality diagnostic services to the population by effective implementation of their model (Table 4).

Table 4. Analysis of IVD models through the enablers component

Enablers					
#	Parameter	Telangana	Tamil Nadu	Chhattisgarh	Odisha
1	Technology – digitalization	The LMIS helps in tracking and reporting results and monitoring of the entire scheme	LMIS supports in managing operations The integrated essential laboratory services (IELS) app for sample transportation	LMIS is currently available at district hubs only; to be expanded to CHC-level hubs	LMIS being utilized for outsourced tests
2	Removing operational barriers	Model stabilized at one to two districts; then scaled up using learnings from the pilots Phased scale up by core central team Optimal staffing ensured through a combination of regular as well as contractual staff Procurement of equipment and consumables are through state medical supplies corporation	Pilot implemented; based on its learnings, a scale up is planned HR filled through regular staffing and contractual modes Procurement streamlined through state agencies for procurement and supplies	Existing infrastructure utilized to reduce capital expenditure at hubs HR pooling done for improving utilization 60–70% of regular staff in facilities, reducing attrition Procurement streamlined through state agencies for procurement and supplies	Strengthening in-house capacity for routine tests Delivery of high-end tests through outsourcing

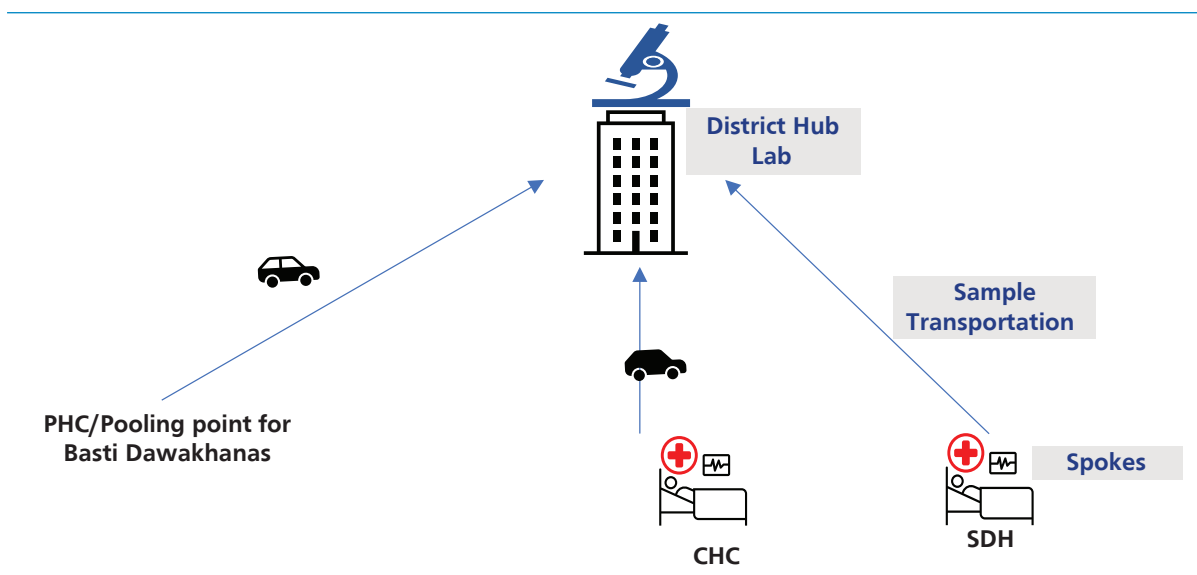
Enablers					
#	Parameter	Telangana	Tamil Nadu	Chhattisgarh	Odisha
3	Policy, governance & financing	Strong political and administrative will Autonomy within department for effective operations Regular reviews and monitoring at state level	Strong political and administrative will Policy of having one MCH per district Merging the scheme with existing flagship schemes	Strong political and administrative will Awareness campaigns run for increasing visibility Administrative focus on improving the quality of tests and scope of services	Strong political and administrative will Outsourced tests to be expanded to PHC-level facilities in the near future

Telangana

Description of the model

Telangana has a functioning in-house diagnostic model, called Telangana Diagnostics or T-Diagnostics, which started in 2018. The programme was initiated in Hyderabad with one hub catering to health facilities in the region. The hub and spoke model is followed, in which samples are collected from each facility, whether PHC, CHC or SDH, as shown in Fig. 3. Samples are then transported to the district hub laboratories through dedicated vehicles for testing and reporting. Currently central laboratories within 31 district hubs across the state provide a total of 134 types of IVD tests. Highly sophisticated equipment is used to manage the load of tests at the district hub central labs. LMIS software is used for smooth functioning of the system. Approximately 52 million tests were conducted in the year 2022 under the T-Diagnostics programme.

Fig. 3. Telangana T-Diagnostics IVD model



Tests provided

A total of 134 tests are provided under the T-Diagnostics scheme.

Sample collection

All PHCs, CHCs, SDHs and DHs act as points for sample collection from patients. Basti Dawakhana (a type of small health facility/clinic known as an urban primary health centre) also serve as collection points, with pooling of samples at PHCs for further transport to the hub laboratory. The samples are obtained from patients by a laboratory technician or staff nurse from 9 am to 1 pm, though the spoke facilities are generally open between 9 am and 4 pm. Samples are barcoded at the collection points and stored in cold boxes, which have been provided to the spoke facilities. As per the norms, a PHC in a rural area caters to a population of 30 000 (in the plains) or 20 000 (in hilly and tribal areas).

Transportation

Dedicated vehicles from an outsourced vendor are used for picking up the samples from spokes and transporting them to hubs. Transportation vehicles have a dedicated route map and cater to a specified set of facilities. The route is designed such that when the sample transportation vehicles start picking up samples, they reach the hub lab within 4 h of the first sample pick up. Each transport vehicle covers eight to twelve facilities, depending on the distances.

Test reporting

Tests are reported for the majority of samples within the same day, with a target maximum of 24 hours. Five per cent of tests are reported after 24 hours. There are plans to reduce the turnaround time further by adopting automation and deploying additional equipment at hubs.

Analysis of the model

The diagnostic model is analysed within the aspects of service delivery, operating model and enablers that contribute to the success of the model.

Service delivery

(a) Service coverage

Primary health centres and higher facilities and 500 Basti Dawakhana are covered in the IVD model. Approximately 4745 SHCs are not covered under the sample transporting programme, but these perform 14 tests through point-of-care (POC) medical devices at SHC facilities.

(b) Turnaround times

Sample pick-up starts after 12 noon and the samples reach the hub around 4 pm. Test results start being reported in the evening of the same day. Results are shared with the patient through phone SMS. Within 12 h, the results of about 95% of the tests are available for printing at the spoke facilities. The printed results are handed over to the patients at the spoke facilities.

(c) Service utilization trends

The model was started in 2018 with a central laboratory in Hyderabad when approximately 3.6 million tests were performed annually. The model expanded to 19 districts of the state in 2021 with new district hubs being established, which increased the uptake to 13 million annual tests. Increased utilization followed, and the number of tests performed was approximately 52 million in the year 2022.

(d) Beneficiary satisfaction

The beneficiaries were satisfied with the scope of the tests available near their homes, which were previously available only in higher-level facilities, requiring more time and travel to access testing. They were satisfied with the minimal waiting time for giving their samples compared to higher-level facilities, which are usually crowded. They were glad that the tests were free of charge, and they were able to receive their reports within a short span of time. Receiving test reports through SMS, eliminating the need for visiting the facilities multiple times, also contributed to their level of satisfaction.

(e) User charges

The tests are free of charge for the beneficiaries.

Operating model

(a) Physical infrastructure at each level

The district hubs are built within separate new buildings in each district hospital complex, outfitted with the latest equipment required for testing. There are currently 31 district hub laboratories functional across the state. Only two districts are yet to have their own district hubs and are being catered to by their neighbouring district hubs.

The spokes are just collection points, each having a small, dedicated area for sample drawing from patients. Each spoke has been provided with two cold boxes for sample transportation.

(b) Operating protocols

SOPs are defined at each level for sample collection, sample transportation, testing and reporting and are displayed at the facilities. The operating protocols guide the flow of tests, as shown in the flowchart below for testing in the greater Hyderabad region.

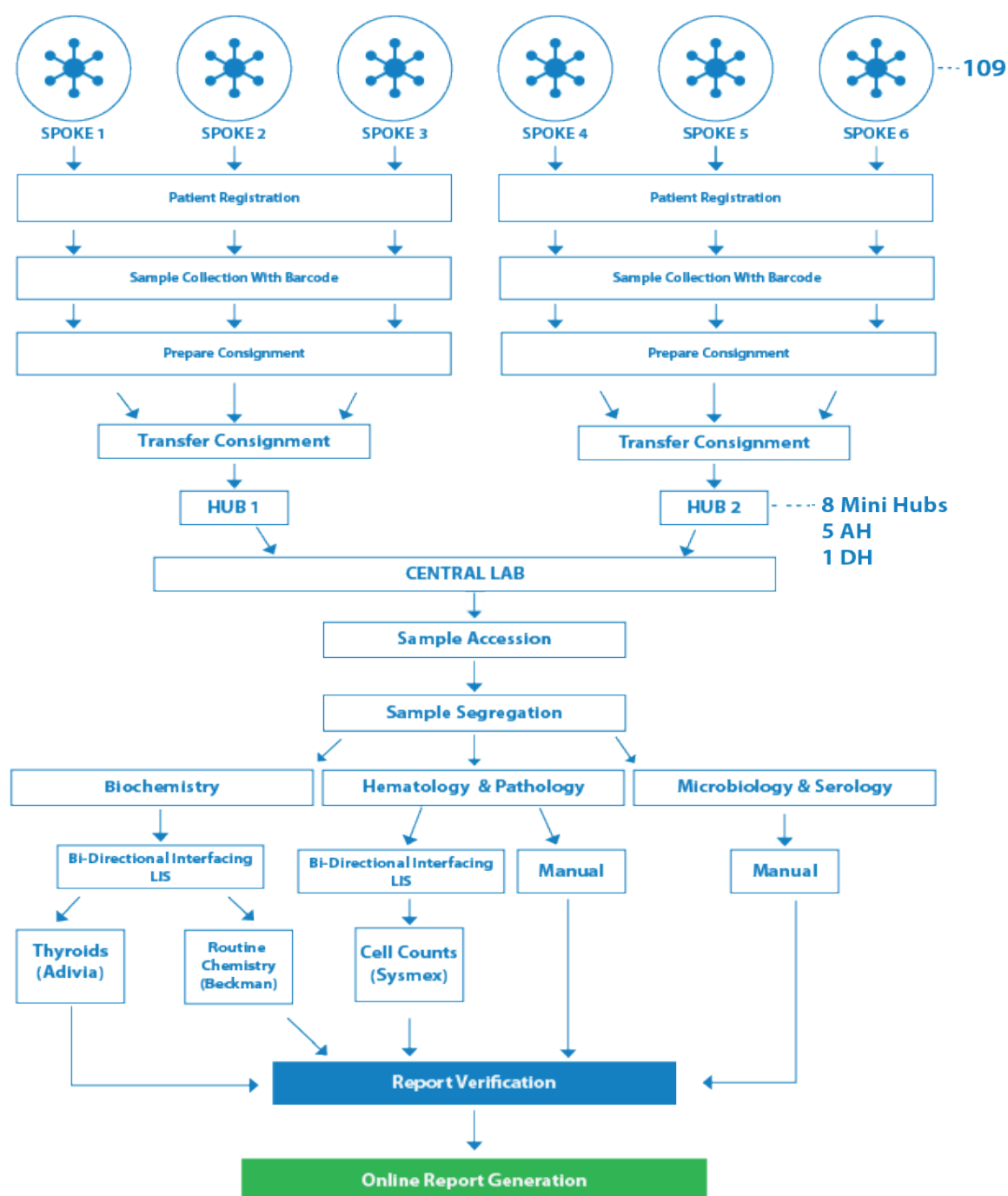
(c) Technology used

T-Diagnostic LMIS software is currently being used and most of the process has been automated. Printed bar codes are provided at the sample collection points, which are affixed to each sample container. The patient and sample details are filled in by the laboratory technician at the spoke before the samples are picked up by the transport agency.

The workflow at the district hub is also automated. After the results are verified by clinical consultants, the test results are put in an SMS queue. Patients get an SMS with a link to the pdf report. A new version of the LMIS is being developed with the Centre for Development of Advanced Computing of India, Ministry of Electronics and Information Technology.

(d) HR capacity

Each spoke facility has a phlebotomist/laboratory technician who collects the samples for subsequent transportation. The district hub laboratory has four clinical consultants, including a pathologist, biochemist, microbiologist and radiologist. They also have a laboratory manager who also acts as a quality manager. Additionally, the hub laboratory has eight laboratory technicians, one data entry operator, one pharmacist, two radiographers and a helper.



Source: <https://tdiagnostics.telangana.gov.in/Scope.aspx>

(e) Quality management system

According to the International Organization for Standardization (ISO) 15189:2012 for medical laboratories, the quality management system present in the diagnostic model was analysed for each of the components below.

SOP and implementation of process control: SOPs for each component – from sample collection to reporting – have been defined at the various facility levels. Process control is ensured based on the SOPs defined. Internal quality control (IQC) for the equipment is done daily and recorded. If the sample load is above 500 per day, IQC tests are run two to three times a day.

Management of documents and records: the LMIS software captures and stores the data from sample collection at the spoke facilities to the reporting of tests from the hubs.

Participation in external quality assessment services (EQAS): this is done for different types of tests through agreements with the All India Institute of Medical Sciences (AIIMS), Delhi and Christian Medical College, Vellore. The hubs receive quality control samples from these institutions every quarter and perform testing to verify the quality of their results.

Physical infrastructure: The district hub is established as a separate structure within the district hospital complex. The physical infrastructure is optimum to manage the load of tests received at the hubs.

Procedures for purchasing and inventory: equipment, reagents and other consumables are purchased through the state-run corporation, Telangana State Medical Services and Infrastructure Development Corporation (TSMSIDC), which is responsible for procuring drugs and medical equipment and related supplies for public health facilities in the state.

Equipment maintenance: calibration of the equipment is done at regular intervals. Maintenance of the equipment is done by vendors/original equipment manufacturers contracted by TSMSIDC.

Customer service: medical officers have conversations with clinical consultants/pathologists and clarify if they have any doubts about the test results. District hub pathologists/consultants also call medical officers to discuss abnormal results of patients when detected.

HR training and capacity-building: regular trainings are provided for the staff posted at hub facilities.

f) Financing

The funds received under the National Health Mission and state government are utilized for the programme.

The cost of establishing a 460 m² hub along with the equipment for pathology, biochemistry, microbiology and radiology for a district with an average population of 1.1 million is approximately US\$ 500 000.

The operating cost of materials at a hub with a load of about 16 000 tests per day is approximately US\$ 20 000 per month for reagents and other consumables.

The sample transportation cost that has been outsourced to vendors in districts is approximately US\$ 1800 per month per route comprising 8–10 spokes.

The maintenance budget for the district hubs is provided to the District Health Society, which is then made available to the District Medical Health Officer for sample transportation and to the superintendent of the district hospital for administrative costs in the district hub laboratories.

g) Monitoring

Data from each hub are collated by the third of each month for monitoring and review. Regular monthly reviews are conducted at the state level. On the fifth of each month, functioning of the IVD model is reviewed at primary care facilities. On the seventh of each month, the performance is monitored of hubs and spokes related to secondary care hospitals. At district-level hubs, the superintendent visits the hubs and conducts weekly reviews.

Enablers

a) Technology

The use of appropriate technology has acted as an enabler in the functioning of this IVD model. The hub-and-spoke model has ensured that the hub has the latest diagnostic equipment for carrying out the requisite tests.

The equipment are able to cater to the entire catchment area, and people from distant places are able to have their samples collected at the spoke facilities, which are tested at the best-in-class infrastructure and technology available at the hub laboratories. This ensures that the same quality of tests available to patients residing within the city limits is also available to patients living far away from cities. The hubs have ensured that the maximum number of tests are available at the primary care level.

A well-designed LMIS ensures that the journey of the sample to the laboratory is tracked, and test results are made available in time to patients in rural and hard-to-reach areas. These results can be accessed from the convenience of their mobiles or in hard copy at the spokes where they provided their samples, which has ensured easy access and availability of services.

b) Removing operational barriers

A key effort has been to remove operational barriers to the functioning of the diagnostic model. The political and administrative will of the government ensured that this programme was well funded and executed.

The laboratory equipment was procured through rate contract holders (vendors) at TSMSIDC, which saved time and streamlined the procurement process. The comprehensive maintenance contract and reagents supply for this equipment are also based on the rates and processes provided by TSMSIDC.

A central core team at the state headquarters took the lead in designing and implementing the hubs and spokes both in terms of the infrastructure and the operations. Roll-out has been done in a phased manner and this central team went from one hub to the next, ensuring that all infrastructure and processes are set up in the district and operations are initiated well. This initial push from the state health department has been critical in ensuring that all infrastructure was set up, commissioned and operations established successfully.

The functioning of the model is regularly reviewed, and directions and guidance are issued from time to time to ensure that the operations are smooth and the model achieves its desired objectives.

c) Policy, governance and financing

There is a strong political and administrative will from the government to establish and operate this model. The management team of T-Diagnostics has complete control over all the hubs, as it has been established as a separate structure within the system. Since the quality of service delivery was prioritized, T-Diagnostics was set up to operate independently for maximal effectiveness and efficiency.

Separate staff have been provided for T-Diagnostics as a mix of deputed staff and contractual staff, which ensures that the hub laboratories function around the clock and seven days a week. Administrative buy-in has ensured that staff available at district level are made available to the hubs. Even those who have been working for specialized programmes in the districts have been made available on a rotational basis, so that services under the IVD model are not disrupted.

Duplication has been avoided in administration. At the district level, the superintendent is the head and controller of the hub. A government order from the health department has ensured that the superintendent of the district hospital is in charge of the hub laboratory. Previously the laboratory was headed by the District Medical Health Officer.

Review meetings have focused on the effectiveness and efficiency of the operations, with special focus on turnaround time. This has ensured that the entire system is geared towards ensuring that the results are delivered in time.

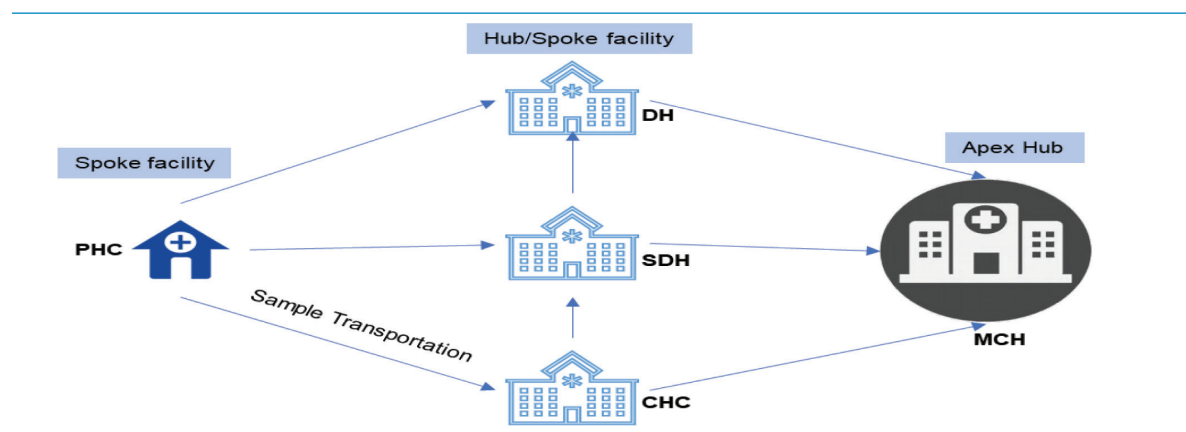
Tamil Nadu

Description of the model

Tamil Nadu is implementing an in-house hub and spoke diagnostic model called Integrated Essential Laboratory Services (IELS). Health facilities including PHCs, CHCs, SDHs, DHs, District Public Health Laboratories (DPHLs) and MCHs are categorized to mandatorily provide an assured number of tests appropriate to their level per the FDSI guidelines. The DPHLs also perform the public health function of surveillance and are steered by the Government of India's guidelines on integrated public health laboratories.

The assured number of tests are either provided through in-house laboratories or through the hub-and-spoke model using sample transportation from lower-level facilities for testing at higher-level facilities. The pilot for the programme was instituted in September 2023 in Tiruvallur district, just west of Chennai. It is planned to be scaled up across the state in 2024 (Fig. 4).

Fig. 4. Tamil Nadu IELS IVD model



Tests provided at each level

A total of 255 tests is provided in the IELS programme. The different facility levels provide a certain number of in-house tests, referring other tests to higher facilities, as shown in Table 5.

Sample collection

In PHCs and CHCs, the samples are obtained from patients from 7 am to 4 pm. The facilities are usually open between 9 am and 4 pm. However, for obtaining fasting blood sugar samples, the timing has been extended to 7 am in the morning. There are cold boxes to store the samples to be transported. Sample collection at the DH and MCH hospitals is round the clock. The samples are obtained from patients by the laboratory technician and staff nurses at the facilities.

Table 5. Tests provided at different levels in the IELS model

Level	In-house tests	Referred to higher facility	Total tests
SHC	14	Nil	14
PHC	29	34	63
CHC	35	62	97
SDH	53	58	111
DH	65	69	134
Non-apex MCH	128	Nil	128
Apex MCH	182	Nil	182
Ultra-apex MCH	255	Nil	255

Transportation

During the pilot implementation, sample transportation from the spokes to the hub was done by the senior treatment supervisor of the tuberculosis (TB) programme using their own transport. The pilot district was divided into 21 clusters, and samples in each cluster were transported by a TB senior treatment supervisor. Each cluster has about five to seven spokes.

The transporter starts at 12 pm from the first facility of the cluster and reaches the final facility by 4 pm. Along the cluster route, the transporter picks up and delivers samples from facilities that can include multiple spokes and hubs. The total distance covered is about 40–50 km.

Engaging transporters for scaling up of the programme is still under process.

Test reporting

Most of the tests were reported within 12–24 hours of sample collection. Some tests, for example, culture, were reported after 24 hours.

Analysis of the model

The diagnostic model is analysed within the aspects of service delivery, operating model and enablers that contribute to the success of the model.

Service delivery

(a) Service coverage

The pilot of the model was conducted in Tiruvallur District for a period of one week during September 2023. It is being expanded across the state in all facilities from PHCs to ultra-apex MCH facilities in the ensuing months. The 14 tests in the SHCs are performed using POC medical devices and do not require sample transportation.

(b) Turnaround times

During the pilot, sample transportation from the facilities started around 12 noon and reached the last facility of the cluster by 4 pm. The test results were updated in the LMIS software within 12–24 hours for most of the samples that were transported. Test results in printed form were available the next day for patients in spoke facilities.

(c) Service utilization trends

During the pilot, approximately 10–20 samples from each primary health care facility like PHCs and CHCs were transported to higher facilities for testing. The PHCs, having an average catchment population of about 33 000, conduct approximately 50 in-house tests daily and CHCs, having an average catchment population of about 150 000, conduct approximately 100 in-house tests daily.

The established in-house laboratories in the CHC, SDH, DH and MCH facilities can act as hubs for the tests transported from lower-level facilities in their catchment areas. Across the state, this includes 37 DHs and 36 MCHs.

MCHs are also proposed for newly formed districts, which can then act as apex hubs for their respective districts. Establishing the service via these existing and planned hub facilities will improve the utilization of diagnostic services once the programme is expanded across the state.

(d) Beneficiary satisfaction

The beneficiaries were glad that the tests conducted at the levels of CHC, SDH, DH and MCH were made available by providing samples near their locality. They were glad that it saved them time and the costs required if they had to give samples for tests at higher-level facilities, which are typically 40–50 km from their locality. The quick reporting time and staff cooperation further increased the level of satisfaction of the beneficiaries.

(e) User charges

There was no registration fee for consultation and all the tests were provided free of charge to the beneficiaries.

Operating model

(a) Physical infrastructure at each level

The central laboratory in the MCH acts as the apex hub for the district. The CHC, SDH and DH facilities act as spokes for the tests to be referred to a higher facility and also act as a hub for their lower-level facilities from which they receive samples for testing. PHCs also have their own laboratory for conducting in-house tests and transport the remaining samples for referral tests to higher facilities. Samples related to malaria and dengue are referred to the DPHL within each district.

Samples for referral tests are placed at pick-up points within each lower facility's in-house laboratory. Designated pouches in five different colours are provided to the spokes for packaging the samples into the cold box for transportation. Each colour indicates the type of destination hub in which those samples will be tested, as shown in Table 6.

Table 6. Colour coding of pouches for sample transportation

Colour of pouch	Type of hub
Yellow	CHC
Green	SDH
Blue	DH
Orange	DPHL
Red	MCH

The equipment, reagents and other consumables were procured through the state-run Tamil Nadu Medical Service Corporation (TNMSC). The computers, dot matrix printers and barcode printers were supplied to all the facilities for the IELS programme, procured through the state government-run Electronics Corporation of Tamil Nadu (ELCOT).

(b) Operating protocols

SOPs have been defined at each level for sample collection, sample transportation, testing of samples and reporting of results.

(c) Technology used

The LMIS software developed by a private vendor is being used for capturing and managing the tests being conducted under the programme. The laboratory technician enters the details of the patient and sample in the LMIS software for both in-house tests and tests to be done at higher-level facilities. The barcode printers are integrated with the LMIS for printing the barcodes for each sample. Test results are updated in the LMIS software, which are printed at the spoke facilities and then distributed to the patients.

A separate web-based IELS application manages sample transportation from the spokes to the hubs. For transport upwards, laboratory technicians at lower-level facilities enter details into the app, assigning samples for pick up to be transported to higher-level facilities. For receiving transported samples, laboratory technicians at higher-level facilities enter associated details into the app.

The IELS mobile application is provided for transporters who undertake the sample transportation process. The mobile application is the extension of the web-based IELS application, and it informs the transporter about the pick-ups and deliveries to be completed. The transporter updates the details at each facility, after completing pick-ups and after completing delivery of samples.

(d) HR capacity

The spoke facilities have a laboratory technician to collect the samples for conducting in-house tests and the samples to be transported to higher-level facilities for subsequent testing. The central laboratory in the MCH has microbiology, pathology and biochemistry consultants along with a biomedical engineer, laboratory technicians and laboratory assistants.

The District Epidemiologist and District Microbiologist have designated roles of monitoring and mentoring in the IELS programme. A dedicated LMIS IT coordinator for each district has been or will be recruited to oversee resolution of hardware and software issues. A biomedical engineer is present at the district level for managing biomedical equipment issues in all the facilities in the district.

(e) Quality management system

As per ISO 15189:2012 for medical laboratories, the quality management system present in the diagnostic model of the state was analysed for each of the components below.

SOP and implementation of process control: SOPs have been defined at each level for the various processes involved in the scheme. IQC tests in the laboratories are run daily by laboratory technicians and the values are plotted on graphs and kept for future reference.

Management of documents and records: the LMIS captures and stores data from sample collection to reporting of tests. Sample transportation is captured through the IELS web and mobile application. There is a plan to integrate the LMIS with the nationwide web-based Health

Monitoring Information System (HMIS) 2.0, which will further streamline documentation of the diagnostic data.

Participation in EQAS: EQAS is done via engagement with Christian Medical College, Vellore, AIIMS, Delhi and Tata Memorial Hospital, Mumbai for various type of tests. The hub laboratories receive samples at regular intervals from these EQAS institutions for conducting quality checks.

Physical infrastructure: the physical infrastructure for diagnostic tests at various levels, from PHC to MCH, is laid out based on the guidelines set by the central and state governments.

Procedures for purchasing and inventory: equipment, reagents and other consumables for diagnostics are procured through TNMSC. IT solutions are procured through ELCOT, which is a state-run electronics corporation.

Equipment maintenance: regular calibration of the equipment is done at defined intervals. Equipment maintenance at the hubs is managed by the biomedical engineer posted at the district level. The MCH has a dedicated biomedical engineer for equipment maintenance. A helpline number of TNMSC is accessible to all the laboratories for resolving any equipment issues.

HR training and capacity-building: regular training for laboratory technicians posted at different levels is conducted at the district level.

(f) Financing

Funds provided by the National Health Mission and state government are utilized for the programme. The funds are utilized for procuring programme reagents and other consumables through TNMSC. Medical officers have been instructed to utilize untied funds for maintaining the equipment related to the IELS, such as dot matrix printers and barcode printers.

(g) Monitoring

During the pilot phase, daily reviews were conducted on the tests done at each level and sample transportation to higher facilities. Statewide training of trainers has been given to the district-level teams for roll-out of the scheme.

The District Epidemiologist and District Microbiologist are responsible for monitoring the IELS and LMIS entries in the district. The LMIS IT coordinator is responsible for receiving feedback from the facilities regarding hardware and software related to the LMIS and IELS applications. Regular reviews at state, district and block levels are planned to be conducted once the programme is rolled out across the state (block-level administration is above the village level and below the subdistrict level).

Enablers

(a) Technology

The LMIS software has been developed to act as the backbone of the scheme and provide a real-time view of the operations. The IELS applications (web-based and mobile) have been developed to ensure that the sample transportation process is smooth. The LMIS is in the process of being integrated with the HMIS 2.0 application.

(b) Removing operational barriers

Existing vacancies of laboratory technicians are being filled by the Medical Recruitment Board. However, separate contractual recruitment of laboratory technicians has been allowed at the MCH based on their requirements, which has improved HR availability.

A block public health laboratory has been proposed by the Central Government Finance Commission, at an estimated cost of approximately US\$ 60 000. This will further strengthen capabilities at the block level in the CHCs.

(c) Policy, governance and financing

The political leadership is focusing on health and education; thus, full-scale roll-out of the programme is being planned.

Establishing one MCH for each district is an important policy decision that has helped to strengthen the in-house capacity of the state to manage the increased load of diagnostic tests at the district level.

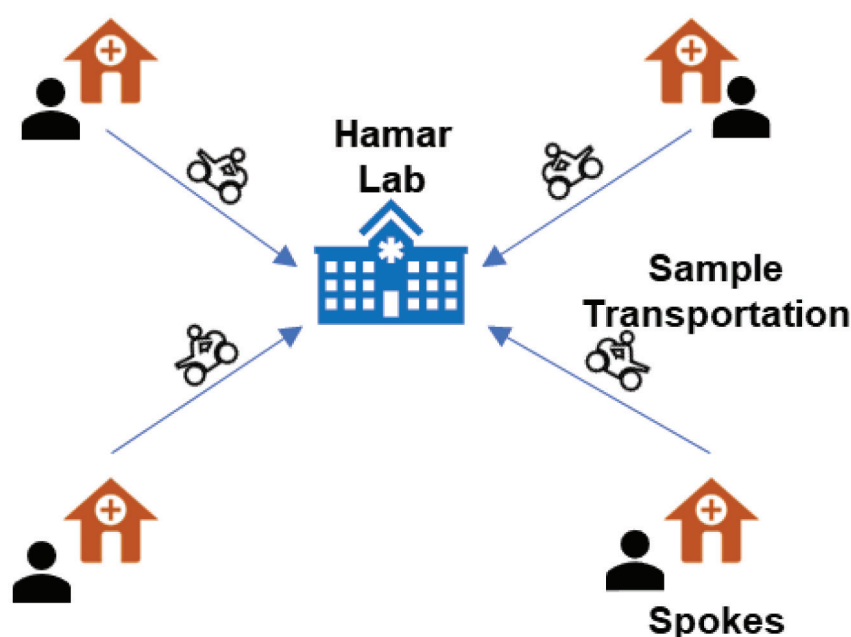
The IELS scheme is planned to become part of the state's "Makkalai Thedi Maruthuvam" (meaning "Health care at doorstep") programme, which is a flagship programme to improve access to health care. This will further improve the reach of the IELS scheme. The in-house capacities of public health facilities will be strengthened by the scheme, which will help ensure sustainability.

Chhattisgarh

Description of the model

A state-driven in-house diagnostic model "Hamar Labs" (meaning "Our Labs") is currently operating in Chhattisgarh, following the hub-and-spoke model. Hubs have been developed at the district hospital level and CHC level to provide state-of-the-art diagnostic services. Currently, Hamar Labs have been set up in 13 DHs and seven CHCs across the state, as well as one MCH (Fig. 5). The state is planning to expand the programme to another 15 DHs and five MCHs in 2024.

Fig. 5. Chhattisgarh IVD model Hamar Labs as Hubs



Tests provided at each level

- At the district hospital level hub, 114 tests are provided, including 66 biochemistry tests, 15 haematology tests, 8 serology tests, 15 pathology tests and 10 microbiology tests.
- At the CHC level hub, 54 tests are provided, including 23 biochemistry tests, 14 haematology tests, 8 serology tests, 6 pathology tests and 3 microbiology tests.
- At the PHC level, 23 in-house tests are conducted.
- At the SHC level, 14 tests are conducted using POC medical devices.

Sample collection

In PHCs and CHCs, sample collection time for in-house tests is from 8 am to 8 pm. For referred tests, samples are collected from patients between 8 am and 2 pm for subsequent transportation. One vaccine box is designated in each PHC and CHC as a cold box for the purpose of sample transportation. Samples from mobile medical units in *haat bazaar* clinics (mobile clinics set up in market areas) are collected by laboratory technicians. Sample collection at DHs is done round the clock by laboratory technicians.

Transportation

Sample transportation is performed by TB runners who each pick up the samples from the three to four spoke facilities in addition to TB sputum samples. The TB runners start sample pick-up at 2 pm and reach the hub by around 5 pm using their own modes of transportation. The typical distance covered by a TB runner is 30–40 km. The samples from the *haat bazaar* clinics are transported by the laboratory technician posted in the mobile medical unit.

Test reporting

Most tests are reported within 12–24 hours of sample collection, with a few tests reported after 24 hours.

Analysis of the model

The diagnostic model is analysed within the aspects of service delivery, operating model and enablers that contribute to the success of the model.

Service delivery

(a) Service coverage

Hamar Labs were initiated in 2020, and now include 13 DH hubs, seven CHC hubs and one MCH hub spread across the state. Expansion is ongoing for another 15 DHs and five MCHs in 2024.

(b) Turnaround times

Sample pick-up from the spoke facilities starts around 2 pm with samples reaching the hub facility by 5 pm. Reports reach the patient within 24 hours of sample collection in the majority of cases.

The TB runner retrieves the printed reports of the previous day's samples while dropping off the current day's samples at a CHC-level hub. At the district-level hub, reports are sent through email to the spoke facilities, which are then printed and distributed to the patients. Reports for samples

from the haat bazaar clinics are retrieved by the laboratory technician the next day for distribution to the patients.

(c) Service utilization trends

Approximately 5–10 samples from each PHC, with an average catchment population of about 35 000, are transported to hub facilities for testing.

During the initial roll-out in 2020, Hamar Lab at Raipur District Hospital, which caters to a population of about two million, tested approximately 250 000 tests annually. This increased to around 700 000 tests from January to October 2023, which extrapolates to 840 000 tests annually, an increase of 3.4 times from 2020 to 2023. The testing volumes are expected to increase when the programme is expanded across the state.

(d) Beneficiary satisfaction

The beneficiaries were satisfied with the advanced tests provided near their locality, eliminating the need to travel long distances to access those tests. The printed reports were available typically the next day, which contributed to the satisfaction of the beneficiaries.

(e) User charges

All the tests are provided free of charge to the beneficiaries.

Operating model

(a) Physical infrastructure

There are two types of hubs present in Hamar Labs: district level and CHC level. The hub laboratories are set up within the existing physical infrastructure, with space of about 230–420 m² for the DHs and 75–140 m² for the CHCs. The equipment, reagents and other consumables are procured through the state-run Chhattisgarh Medical Service Corporation (CGMSC).

PHCs have their own laboratory for conducting in-house tests and transport the remaining samples for referral testing at the Hamar Labs. Samples for the referral tests are gathered at pick-up points within their in-house laboratory. A designated vaccine box is used as a cold box for transportation of samples to the hubs.

(b) Operating protocols

SOPs have been defined at each level for sample collection, sample transportation, testing of samples and reporting results.

(c) Technology used

Currently, the LMIS software is only operational in 13 district-level Hamar Labs. Barcoding is done for the samples collected in the DHs, but not for the samples being transported from the spoke facilities. Details are written manually for samples collected in the spoke facilities.

In district-level hubs, sample details from the spokes are entered into the LMIS system and final reports are communicated through email to the spoke facilities. The reports are printed at the spokes and distributed to the patients. Printed test reports are also sent via the TB runner when they drop off samples.

Emergency and critical value results are immediately communicated to the concerned doctor by telephone. An SMS service for report dispatching was briefly implemented, but it will be fully operational when the LMIS is implemented completely.

(d) HR capacity

The spoke facilities have laboratory technicians to collect the samples for conducting in-house tests and the samples to be transported to the Hamar Lab at a higher-level facility for subsequent testing. The DH hub laboratory has microbiology, pathology and biochemistry consultants along with nine laboratory technicians and three laboratory assistants. The CHC level hub laboratory has medical officers along with eight laboratory technicians and two laboratory assistants. A biomedical engineer is available on call from the private vendor engaged by CGMSC for the maintenance of biomedical equipment.

(e) Quality management system

As per ISO 15189:2012 for medical laboratories, the quality management system present in the diagnostic model of the state was analysed for each of the components below.

SOP and implementation of process control: SOPs were defined for each process from sample collection to reporting at different levels. IQC tests are run daily at the start by the laboratory technicians with values recorded for future reference.

Management of documents and records: the LMIS is used to capture the process from sample collection to reporting at the district-level hub and their respective spoke facilities. Manual records are used to capture the process at the CHC-level hub and the respective spoke facilities.

Participation in EQAS: EQAS is done by engaging with Christian Medical College, Vellore and AIIMS, Delhi for various type of tests. The test controls for each of the equipment are procured through CGMSC.

Physical infrastructure: the existing physical infrastructure at DHs and CHCs are used to establish the hub laboratories. The physical infrastructure is sufficient to cater to the case load.

Procedures for purchasing and inventory: equipment, reagents and other consumables are procured through CGMSC.

Equipment maintenance: the equipment is maintained by the private vendor identified by CGMSC. A toll-free helpline number of the vendor is provided to all the laboratories for resolving issues regarding the maintenance of biomedical equipment.

(f) Financing

Funds under the National Health Mission and state government are utilized for the programme.

The district-level hub requires space of about 230–420 m² and the cost of setting up the infrastructure is approximately US\$ 60 000–110 000. A district-level hub caters to an average population of 1 million.

The CHC-level hub with a space of 75–140 m² has an infrastructure set-up cost of approximately US\$ 20 000–35 000.

Each district has been allotted approximately US\$ 6000–12 000 of untied funds to procure reagents and other consumables during emergencies and stock-outs.

The TB runner is currently being paid a fuel allowance of about US\$ 0.08/km for sample transportation. An additional incentive of about US\$ 2/day has been proposed.

(g) Monitoring

Service utilization of 13 district-level hubs is regularly monitored by state-level officials using their LMIS dashboard. The utilization of the remaining hubs is monitored through the data captured regularly through manual reports and Google forms.

Regular district-level monitoring of Hamar Labs is done by the Chief Medical Health Officer. Monthly review of Hamar Labs along with other health-care initiatives is done by the Chief Secretary. Quarterly review of Hamar Labs along with other programmes is also done by the health minister of the state.

Enablers

(a) Technology

The LMIS software is currently in the process of being streamlined and implemented across the state for smooth functioning of Hamar Labs. This software supports functional operation and aids in getting a near real-time picture of the functioning of the scheme.

(b) Removing operational barriers

Financing and HR-related issues have been addressed to improve the effectiveness and efficiency of the scheme. Vacancies of laboratory technicians at both hub and spoke levels of facilities have been filled through regular and contractual appointments. To operate Hamar Labs, existing laboratory technicians have been pooled from various vertical programmes, including the Integrated Counselling and Testing Centre, Noncommunicable Diseases Programme, National Health Programme for Children (Rashtriya Bal Swasthya Karyakram), among others.

Existing physical infrastructure is used to set up the hub laboratories, which reduces the capital expenditure for establishing the laboratories.

c) Policy, governance and financing

The political leadership of the state has focused on the agriculture, health and education sectors to maintain goodwill among the public, which has helped to expand the programme across the state. Along with Hamar Clinics (Hamar Clinic meaning "Our Clinic"), awareness campaigns about Hamar Labs are carried out across the state.

About 60–70% of Hamar Labs staff are regular employees, which aids in the stability of the programme; the remaining are contractual or temporary staff. HR resource pooling done at the hubs improves effective utilization of resources.

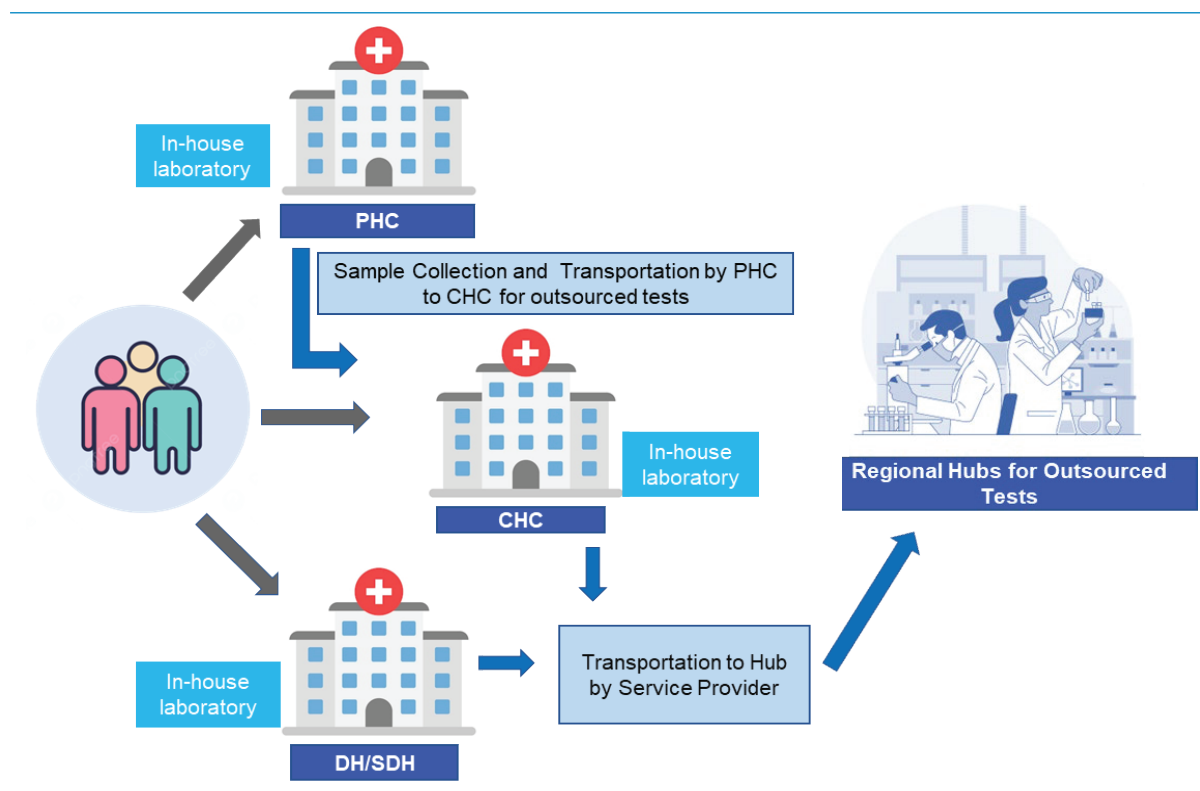
As Hamar Labs are developed within the health system, in-house capacities of public health facilities are strengthened, which helps in the sustainability of the programme. The current focus of the leadership is to ensure the quality of the tests performed by Hamar Labs for effective diagnosis and treatment. They further focus on improving the scope of services being provided, in line with the FDSI guidelines.

Odisha

Description of the model

The Government of Odisha launched “Free Diagnostic Services–Nidaan” (meaning “Treatment”) with the vision of providing accessible and affordable quality health-care services to the citizens. The initiative aims to reduce out-of-pocket expenditure of beneficiaries for diagnostics via provision of these at public health facilities, and it contributes to the attainment of UHC and the Sustainable Development Goals. This initiative was launched by the Honourable Chief Minister of Odisha on 1 January 2018, starting with strengthening district-level institutions. Scaling up started in 2023 with the reach extended to peripheral health facilities in a hybrid hub-and-spoke model: strengthening in-house capacity of the existing public health facilities and engaging a private service provider for high-end tests, as shown in Fig. 6.

Fig. 6. Odisha IVD model



Tests provided at each level

There are 146 tests provided at different levels: DHs provide 146 tests, SDHs provide 121 tests, CHCs provide 103 tests, PHCs provide 65 tests and SHCs provide 16 tests. Among the 146 tests, 52 tests are outsourced to a private service provider in a PPP.

Sample collection

The government has provisioned deployment of a dedicated laboratory technician at PHC level for sample collection and performing in-house tests. Existing nursing personnel posted at the SHC level manage their in-house testing using POC medical devices.

Transportation

The PHC is responsible for arranging transportation of outsourced tests to the nearest CHC along with completion of a test request form. For the samples received from the PHC, the tests covered under the in-house tests of the CHC are performed at the CHC laboratory. The samples to be further outsourced are barcoded in the CHC. The private service provider picks up samples for testing from every CHC or DH/SDH spoke daily.

Test reporting

Most tests that are conducted at the in-house laboratories are reported within 12–24 h. For outsourced tests, reporting is done based on agreed timelines, which varies from one day to eight days based on the type of test.

Analysis of the model

The diagnostic model is analysed within the aspects of service delivery, operating model and enablers that contribute to the success of the model.

Service delivery

(a) Service coverage

All the health facilities are mandated to provide diagnostics services according to the Indian NEDL. Outsourced testing facilities have been provisioned along with in-house laboratory facilities. Currently, outsourced laboratory facilities are available for samples picked up from the CHC level and above. In-house laboratory facilities are operational at PHCs where laboratory technicians are posted. Planning is in process for outsourcing laboratory services at the PHC level to fill in gaps.

(b) Turnaround times

Sample collection takes place during outpatient department hours. In-house test results are provided to the beneficiaries through printed test result reports. Outsourced testing results are shared through SMS with a hyperlink to download the reports.

Spokes have been given online access to track the progress of testing and download the reports for beneficiaries. Health facilities also maintain a register to record the test results for the offline system of reporting.

(c) Service utilization trends

At the PHC level, with a typical catchment population of about 30 000, about 8–10 beneficiaries are prescribed laboratory tests daily.

At the CHC level, with a typical catchment population of about 130 000, about 20–25 beneficiaries are prescribed laboratory tests daily. For outsourced laboratory tests, CHCs produce about 62% of the workload, followed by DHs at 28% and SDHs at 9%.

(d) Beneficiary satisfaction

The beneficiaries were glad that the tests are being performed free of charge. They were happy that even high-end tests were accessible near them. Typically, beneficiaries would earlier have to travel 30–40 km to get tested. Quick turnaround times and delivery of test results through SMS has further improved the satisfaction levels of the beneficiaries.

(e) User charges

These tests are provided free of charge to the beneficiaries.

Operating model

(a) Physical infrastructure at each level

The government is responsible for setting up each in-house “integrated laboratory”, including equipment. The private service provider is responsible for the establishment and management of the six regional hub laboratories across the state. The private service provider is also responsible for sample pick-up from the level of the CHC up to the DH, subsequent transportation and testing.

(b) Operating protocols

SOPs are defined at each level – for sample collection, sample transportation, testing and reporting.

(c) Technology used

In-house facilities work offline with manual reports given to the patients. The LMIS software is used by the private service provider for outsourced tests, which captures the testing process from sample pick-up to report delivery. Barcoding of samples is done for outsourced tests and SMS test results are provided to the patients undergoing outsourced tests.

(d) HR capacity

Each integrated laboratory down to the PHC level has provision for at least one laboratory technician, which increases depending on the load at the higher facility levels. For outsourced tests, there are six regional hubs present across the state, with one hub functioning as an apex hub, which manages logistics and supplies to the other five hubs. The apex hub has 60–65 personnel, while the remaining five hubs have 30–35 personnel each.

(e) Quality management system

As per ISO 15189:2012 for medical laboratories, the quality management system present in the diagnostic model of the state was analysed for each of the components below.

SOP and implementation of process control: SOPs for service delivery were defined at each level. IQC was done regularly and recorded.

Management of documents and records: the LMIS is used to capture the process followed in the outsourced tests, from sample pick-up to reporting. Manual records are maintained.

Participation in EQAS: EQAS of diagnostic tests is done by a private vendor engaged through the Odisha State Medical Corporation (OSMC), which is a state-run medical service corporation.

Physical infrastructure: the in-house laboratories are established at the PHC, CHC and DH levels. Six regional laboratories have been set up across the state by the private service provider for outsourced tests. The physical infrastructure is sufficient to cater for the case load.

Procedures for purchasing and inventory: equipment, reagents and other consumables are procured through OSMC.

Equipment maintenance: in-house equipment are maintained by a vendor identified through OSMC.

Customer service: customer service support is provided by the private service provider for the outsourced tests. Dedicated customer support staff are present at the regional laboratories to address the queries of patients regarding their tests.

(f) Financing

The programme is funded by the government through central and state contributions, along with funds from the Central Government Finance Commission.

(g) Monitoring

The health departments at state, district and block levels review the progress of the programme at regular intervals as part of other existing programme reviews.

Enablers

(a) Technology

The hybrid model augments the availability of a wider range of tests to the population than are typically provided with in-house models. Outsourced testing for higher-end IVDs is available via higher technology equipment at the private service provider's regional laboratories.

The LMIS software is used by the service provider for managing the outsourced tests, which captures the complete workflow, from sample pick-up to test report delivery. Barcoding of samples is done for outsourced tests. Patients receive their results through SMS for the outsourced tests.

(b) Removing operational barriers

The Indian NEDL has been implemented for all levels of health facilities. Infrastructure as well as HR deployment have been focused on to enable diagnostics services at all levels of health facilities. The quality of services is ensured through regular capacity-building and adherence to the SOPs.

(c) Policy, governance and financing

The scheme was launched by the Honourable Chief Minister of Odisha, which shows the strong political and administrative will present in the state. The government has prioritized the initiative down to SHC-level health facilities, with the provision for outsourced referral tests down to PHC level (expected to get fully operationalized in early- to mid-2024). The government has selected the hybrid model to strengthen in-house capacities along with outsourcing to enable delivery of high-end services to patients.

Salient features of the IVD models

The implementation of the IVD models in all the states has resulted in improved availability, accessibility and affordability of diagnostics at the primary care level. These states have implemented a hub-and-spoke model either completely in-house or through a hybrid approach of in-house plus support from a private service provider. The reason for differences in the models are the variations in each state's specific geographical challenges, availability of financial resources, availability of HR and acceptance by health-care workers. While there are common strands within the models, there are also certain unique features among the states.

Common features include the following

- (1) **Aggregation of demand:** a hub-and-spoke model is an effective way of aggregating demand from multiple primary care facilities. The aggregated demand is then sufficient to feed the diagnostic equipment at hub facilities, which are typically at a subdistrict cluster, district or regional level, depending on the design. This ensures efficient utilization of equipment at the hub facility, which promotes cost-effectiveness. However, success is heavily dependent on a robust sample transportation network, which must be done in parallel with strengthening in-house services. Thus, basic tests are done in-house at the appropriate facility level, with higher-end tests transported up to higher facilities.
- (2) **Managing HR capability requirements effectively across the value chain:** primary care facilities act as spokes and just require a laboratory technician/nurse/phlebotomist to collect samples from patients. Transportation can be done by nontechnical personnel using cold boxes and a well-defined SOP to ensure the integrity of the samples. The hubs are usually at a higher-level facility and require technically qualified staff, such as laboratory technicians, clinical specialists and biomedical engineers. Establishing laboratory-qualified HR at cluster/district/regional levels is much easier than placing specialized HR in primary care facilities.
- (3) **Use of technology:** Availability of high end equipment at the hub labs and establishing effective maintenance are prerequisite for model functionality. Such high-end equipment is fully automated, thereby averting possible manual errors in the whole process. The different models ensure that the high-end equipment is serviceable effectively, being deployed at cluster, district or regional level. LMIS software enables tracking of each sample from the point of collection through testing until reporting, including sending reports directly to beneficiaries through SMS with a link to download the reports.
- (4) **Political and administrative will:** in all the states, there has been a strong political and administrative will to implement the scheme effectively. This has ensured adequate funding and allowed operational bottlenecks to be solved in a timely manner. Effective use of existing resources within the entire health system ensured that operations were not impeded by a shortfall of infrastructure, HR or finances. To support effective purchase of equipment and reagents and maintenance of the equipment, the state medical equipment and drugs procurement agency empanelled suppliers, prequalifying the requisite goods and services.
- (5) **Effective monitoring:** schemes are monitored at multiple levels, including district/region/state levels. Regular monitoring is supported by data from the LMIS, which has assisted in highlighting issues of concern, leading to course corrections and learnings.
- (6) **Standardized procurement:** procurement of equipment, reagents and other consumables is done by the state-run medical service corporations. Concentrating procurement services at the upper level helps in ensuring low prices and steady supplies.

Some unique features of different models among the states are highlighted below.

Telangana

- The T-Diagnostics service was established as a separate structure within the system. Its autonomy has helped ensure effective functioning.
- State-of-the-art equipment was procured for the programme for efficient management of case load.

- Delays in sample transportation from the spokes to the district hubs were minimized through the use of dedicated vehicles from an outsourced vendor.
- Phased implementation of the programme ensured the development of solutions to bottlenecks uncovered during the pilot and subsequent early experiences.
- The use of technology such as LMIS and automation of the district hub laboratory has ensured short test turnaround times.

Tamil Nadu

- A main objective of the Tamil Nadu IELS is to strengthen the in-house laboratory capacities of health facilities at different levels, from the PHC to the MCH, with tests done through in-house laboratories and referral to higher facilities.
- Colour coding of samples based on the destination hub laboratory avoids mistakes during transportation, providing guidance to the transporter for delivering samples to the correctly designated hubs.
- The use of LMIS and the IELS applications support quality control of the entire process, from sample collection to test reporting.
- IT solutions are procured from the state-run ELCOT, which provides direct support.

Chhattisgarh

- Laboratories are being established within the present facilities at district and CHC levels, which helps to strengthen the in-house capacities of health facilities. This helps to ensure sustainability of the programme.
- HR optimization was performed by HR pooling, utilizing laboratory technicians from multiple vertical programmes for testing samples at the hub laboratory.
- Providing an honorarium of US\$ 2/day in addition to the petrol allowance of ~US\$ 0.08/km may help reduce the attrition rate of transporters.

Odisha

- The hybrid model ensures strengthening of the in-house capacities of health-care facilities, while also widening the scope of services through outsourcing of advanced tests not available in-house.

Pros and cons of different IVD models

The IVD models follow either an in-house model or a hybrid model that adds PPP to in-house diagnostics services. A few strengths and weaknesses were observed, which could point to further strengthening of the models.

In-house model

Pros

- It strengthens in-house service delivery capabilities over time.
- Procurement through state-run medical service corporations helps to streamline the supply chain, ensuring steady supplies at reasonable rates.

- Barcoding of samples at the spoke facilities improves the process of sample transportation and helps to minimize incorrect reporting.
- Having dedicated infrastructure, HR, IT solutions and sample transportation modalities ensures effective service delivery and enhances accountability.

Cons

- Lack of LMIS that extends across all levels to capture the entire process inhibits effective monitoring mechanisms, which can lead to poor quality of services being provided.
- Variable quality of services could be expected, with higher-level facilities following quality standards due to more frequent and effective monitoring, which is not the case for lower-level facilities located more remotely.
- Utilizing existing HR for sample transportation increases their workload, which could adversely affect the quality of services provided.
- Sharing of resources with other programmes may lead to less effective service delivery and reduced accountability, if those resources are spread too thin.

Hybrid model (in-house + outsourced)

Pros

- Cost-effective way of delivering high-end tests with low volumes for a large population. This can help governments focus on improving the capabilities of in-house laboratories for providing low-cost high-volume tests.
- Providing high-end tests via the private sector enables coverage of the entire population with those tests. This can lead to lower costs compared to direct private sector (market) rates for testing.
- Sample transportation through the private service provider reduces government workload, enabling focus on strengthening in-house laboratory services.

Cons

- The cost of delivering high-end testing services may be on the higher side, compared to the situation where in-house capabilities are sufficient to provide these high-end services.
- Lack of a strong monitoring and review mechanism could result in deterioration in the quality of the services provided.

Conclusion – IVD models

The IVD models were captured from four different Indian states with varying degrees of existing infrastructure and resources. Each model has its nuances, depending on the local context. Although these are early experiences observed from the states, all four models appear to effectively deliver the intended services. To improve access at primary health care facilities, states have implemented effective sample transportation mechanisms, implemented IT systems for streamlining testing and reporting, judiciously utilized HR resources, aggregated demand, leveraged existing effective procurement systems, relied on strong political will, instituted effective monitoring and collaborated with private entities where applicable.

3. Teleradiology models

Teleradiology

General overview

Teleradiology is defined as the transmission of radiological images of patients from investigations such as X-rays, CT scan and MRI from one location to another, with the images analysed by a radiologist or other specialist at a distant location, followed by reporting back to the originating facility.

Teleradiology at the primary care level is considered an important step towards achieving UHC, as it enables patients at distant locations to access diagnostic imaging, previously not feasible due to the limited availability of skilled clinicians. Accessibility of diagnostics imaging services at lower-level facilities improves the quality of clinical services and reduces the time burden and out-of-pocket expenditure of patients. The Indian Public Health Standards recommend X-ray and ultrasound services in health-care facilities from the CHC level and above.

Technology involved

Teleradiology is made possible by a combination of hardware and software technology, applied to the images collected and transmitted and to the reporting of the tests by a remote radiologist.

The computed radiography (CR) system is the basic hardware required for telereporting of X-rays for diagnosis. The imaging plates can be erased and reused, a major advantage over conventional X-ray plates. There is a potential reduction in exposure to radiation and processing time while using CR systems. ISO certification and regulatory authority approval must be present to ensure the safety and effectiveness of the equipment. CR systems are the preferred alternative to digitizers used with conventional X-ray equipment, which can digitally scan X-rays.

The PACS software is used for storage and digital transmission of the radiological images. PACS eliminates the need for physically storing the films and reports; instead, securely storing these in the servers that are retrievable remotely from multiple locations. Radiologists in remote locations can interpret and report on the radiological investigations received from multiple spoke facilities.

Regulations must be followed while storing and transferring health-care information, for security of patient confidentiality. Health Level Seven (HL7) accredited standards are to be followed for sharing, exchange, retrieval and integration of health information used for clinical purposes and delivery of health-care services. HL7 refers to the seventh layer of the ISO seven-layer communications model for Open Systems Interconnection. The United States Health Insurance Portability and Accountability Act mandated certain standards for protecting sensitive patient information from being revealed without the consent and knowledge of patients. These standards provide the model for informed consent obtained from the patient availing teleradiology services.

Teleradiology in India

Teleradiology was included as an important component of the FDSI, because patient out-of-pocket expenditure is significantly higher per test for radiological investigations when seeking care outside the public sector, compared to IVD laboratory tests.

Three hub-and-spoke models were proposed to states for providing services through PPP, depending on the degree of involvement of the private service provider.

- Model 1: the service provider is responsible for the HR, equipment and telereporting by radiologists. The states provide the space and electricity for setting up the spoke facility.
- Model 2: the service provider is responsible for the HR and telereporting by radiologists. The states own the equipment and provide space and electricity for the spoke facilities.
- Model 3: the service provider is responsible for telereporting by radiologists. The states provide the equipment, space and electricity, and HR.

Currently, 11 states in India have established a functioning teleradiology model in a hub-and-spoke approach. The National Health Mission and respective state governments provide the required funding. The spoke facilities are usually set up from CHC-level facilities and above. Turnaround time for test reporting is typically within 12 hours from the time of imaging and within six hours for emergency cases.

Radiologists, who interpret the images sent through PACS and report back the results to the spoke facilities, are located in hubs that are typically established at the state capitals. In some cases, the telereporting radiologist is located outside the state where the spokes conduct imaging tests. The printed reports are distributed at the spoke facilities. There is also a provision for receiving the reports through SMS in a few states.

Case study: Telangana

Description of the model

Telangana has a teleradiology service in hub and spoke model functioning in the greater Hyderabad region. Mini hubs are set up at the facilities above the level of PHCs, which provide ultrasonography (USG) and X-ray services, including mammography. These are termed “mini hubs”, because they aggregate patients at a mid-level facility, e.g., SDH, from several lower-level facilities, e.g., CHCs and PHCs. However, the radiologist is located at different facility, the hub. Recently, a radiologist has been placed at a former mini hub, effectively converting it to the hub for all the mini hubs. Previously, Nizam Medical College acted as the hub in the region. PACS software is used for image transmission and reporting purpose. Expansion of the model to other regions in the near future is being planned.

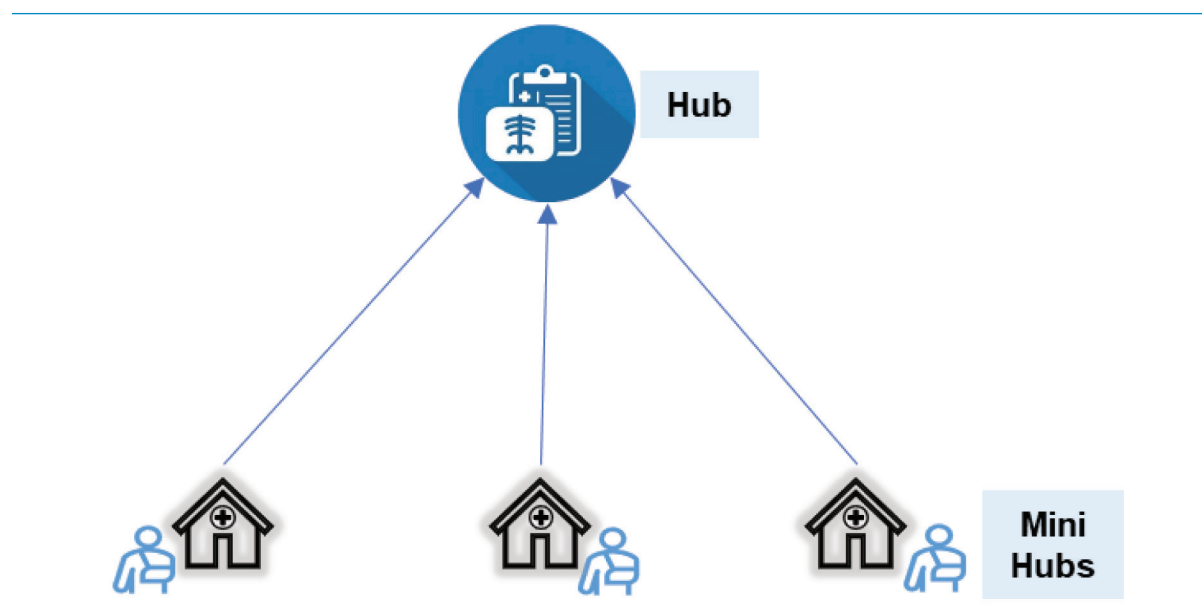
Tests provided and reporting

Mini hubs are currently operational in the greater Hyderabad region at the level of the SDH. They provide X-ray, USG and mammography services to the patients. Technicians conduct the tests and share the images with the hub through the PACS software. The radiologist is based at the hub for interpreting the images sent from all the mini hubs and reporting the results. The reports are usually delivered within 24 hours to the spoke facilities after being analysed by the hub radiologist.

Analysis of the model

The teleradiology model is analysed within the aspects of service delivery, operating model and enablers that contribute to the success of the model.

Fig. 7. Telangana teleradiology model



(a) Service delivery

(i) Service coverage

The PHCs and SHCs in the greater Hyderabad region are catered by the mini hubs at SDHs. X-ray, USG and mammography services are provided through the mini hubs to the patients, who travel to the SDH for imaging. Patients at the primary care facilities find it convenient to approach a mini hub which is closer to the community and has shorter waiting times as compared to bigger hospitals in the city.

(ii) Turnaround times

Imaging tests are conducted at the mini hubs by the technicians, who share the images with the hub on the same day using the PACS software. The hub radiologist reviews the images and provides reports on the scans within 24 hours. The reports are sent to the mini hubs through the software, which are then printed and distributed to the patients.

(iii) Service utilization trends

As a highly populated region, the service aggregates demand from PHCs and SHCs and passes it on to SDH mini hubs, with associated relief for higher-level facilities. The utilization of services is expected to increase once it is expanded to other regions.

(iv) Beneficiary satisfaction

The beneficiaries were satisfied with the various types of services provided at the mini hubs, such as electrocardiography in addition to imaging tests. The mini hubs were easily accessible to the beneficiaries and had less waiting time compared to higher-level facilities, which improved their satisfaction. They were glad that these services were provided free of charge, which otherwise produces significant costs in the private sector. The turnaround time for receiving tests were within a day, which improved the satisfaction of beneficiaries.

(v) User charges

These tests are provided free of charge to the beneficiaries.

(b) Operating model

(i) Physical infrastructure at each level

Mini hubs are set up in a separate building within the health facilities. The equipment for X-ray, USG and mammography are provided to the mini hubs. The hub has an in-position radiologist and the equipment required for reporting the test results.

(ii) Operating protocols

SOPs are defined at mini hubs and the hub for testing and reporting, and are displayed at the facilities.

(iii) Technology used

The PACS software is used for image transmission from the mini hubs and for reporting. The latest imaging equipment has been procured for the mini hubs. The equipment for printing the reports and scans is present at the mini hubs.

(iv) HR capacity

Each mini hub has a technician to conduct the tests and send the images for reporting. For USG, a doctor (radiologist/sonologist) visits the mini hubs on a rotation basis to conduct USG tests. The hub has the radiologist review images and provide reports.

(v) Quality and safety

IQC and calibration of equipment are done regularly.

(vi) Financing

The government funds the programme.

(vii) Monitoring

Regular review meetings are undertaken by the government to monitor the services provided. Regular reviews are also conducted of the infrastructure and equipment.

(c) Enablers

(i) Technology

The PACS software is used for image transmission and reporting. Equipment for printing reports and scans is provisioned at each mini hub.

(ii) Removing operational barriers

Separate infrastructure has been developed for the mini hubs in the health facilities, which enables them to act independently. The presence of mini hubs has reduced the load on higher-level health facilities, improving the quality of health care being delivered through optimal utilization. The quality and safety of services are ensured through regular capacity-building and adherence to SOPs. The diagnostic mini hub concept at the block level has brought diagnostic interventions closer to the community, which reduces out-of-pocket expenditure. The model has improved the access to high-end diagnostics for the population in India.

(iii) Policy, governance and financing

The scheme is currently functional in the greater Hyderabad region, with plans for expansion to other regions due to the strong political and administrative will to improve the range of services provided to the public. A memorandum of understanding was earlier signed with a government medical college in Hyderabad for reporting of the tests conducted in the mini hubs. Currently, the government has converted one of the mini hubs into a hub by placing a radiologist to perform the image interpretation and reporting functions, which further shows their commitment to strengthening in-house capacity for service delivery.

4. Conclusion

In 2019, India developed and published the world's first NEDL, which built upon the initial list created with the FDSI in 2015. Several Indian states have successfully implemented the provision of essential diagnostics services at all levels of health-care facilities. To improve access to diagnostics for primary health care, states have implemented effective sample transportation mechanisms, implemented IT systems for streamlining testing and reporting, judiciously utilized HR resources, aggregated demand, leveraged existing effective procurement systems, relied on strong political will, instituted effective monitoring, and collaborated with private entities where applicable.

Results showed that NEDL implementation significantly improves access to diagnostics. The state of Telangana saw a large increase in IVD test utilization over four years, from 3.6 million tests at one hub laboratory in 2018 prior to NEDL implementation, to 52 million tests in 31 hub laboratories in 2022. In the state of Chhattisgarh, where implementation started only in 2020, a large DH testing hub saw a threefold increase in IVD test utilization from 33 000 tests per month in 2020 to 100 000 tests per month in 2023. All four states established access to IVD tests at facilities without laboratories. IVD testing went from zero to between five and 20 IVD tests per day per primary health-care facility, enabled via sample transport to hub testing facilities. Across the states, the range of IVD tests available is 134–146 tests. Results are available within 12–24 hours for the majority of tests, with results for some high-end tests taking one to eight days.

Teleradiology services have been established in 11 states in India, providing access to imaging services at lower-level facilities that lack a specialist such as a radiologist for image analysis. These services include X-ray (including mammography), CT scan and MRI.

Although beyond the scope of this study, the dramatically improved access to and utilization of IVD and imaging tests has certainly improved health outcomes and reduced out-of-pocket expenditure. Other positive effects can be assumed, such as reduced antimicrobial resistance and gains in efficiencies by pushing health-care service utilization down to lower-level facilities.

Improving access to clinical diagnostics can be achieved by the creation of a national essential diagnostics list, followed by a staged process to implement it. Few WHO Member States have experience in implementing a national essential diagnostics list, especially at the primary care level.

This study examines the different features of diagnostics service delivery using sample transport for in vitro diagnostics to achieve access at primary health care facilities in four different states in India. It also examined the implementation of teleradiology services to enable access to imaging tests at lower-level health-care facilities that lack specialists for interpreting and reporting on the images.

